ANATOMY OF AI AGENTS 2026

www.arianprabowo.com · 1. Silicon → 10. Swarms × 4 Organisms

① COLUMNS (1 → 10): 1. Silicon → 2. Python → 3. Data → 4. Pre-Train → 5. SFT → 6. Post-Train → 7. Test-Time → 8. Context → 9. Harness → 10. Swarms
② ROWS (4 ORGANISMS): Core LLM Spine + Separated RAG, Multimodal & Embodied Organisms
③ INSIDE EVERY CELL: Flagship Pill (H2) + Checkpoints (H3) + Core Stack & Red ⚠ ANTI-PATTERNS Box (H4)

*Synthesized from Haggai Roitman's 'The Hitchhiker's Guide to Agentic AI: From Foundations to Systems' (arXiv:2606.24937, v1.4, June 2026) + 2026 systems tooling · Companion to AI Skill Tree 2026 & AI Ladder Evolution by Arian Prabowo · Updated: 2026-11-08 02:29:57 AEDT (UTC+11:00)

ANATOMY 1 → 10 Stages LLM · RAG · Multimodal · Embodied (+ ⚠ Traps)	**1. Silicon** Physical Compute Silicon, GPU Clusters, HBM3e Memory & Interconnect Fabrics • Unit: FLOPs, HBM Bytes & Warps • Guide: Ch. 1.6, 2.3, 11.10-14, 30.5	**2. Python** Python Runtimes, Package Managers, Virtual Environments & Tensor Compilers • Unit: Wheels, Tensors & Paged KV • Guide: Ch. 1.24-25, 11.1-9, 21.5, 22.1	**3. Data** Web & Code Corpora, Subword Tokenization & Curation Pipelines • Unit: Corpora, Subwords & Shards • Guide: Ch. 1.2, 3.1, 3.3-3.5, 4.1, 30.2	**4. Pre-Train** Foundation Model Architectures, Sparse MoE Routing & Self-Supervised Pre-Training • Unit: Transformer Blocks & MoE • Guide: Ch. 1.1, 2.2, 3.1, 11.1-9, 30.1	**5. SFT** Supervised Fine-Tuning, Instruction Templates, Sequence Packing & LoRA Adapters • Unit: ChatML, Packing & LoRA • Guide: Ch. 1.8, 3.2, 7.6, 9.4, 30.2
CORE SPINE **LLM** Autoregressive Text, Code & Reasoning Language Models Token-by-Token Language, Symbolic Math, Code Gen & Autonomous Agent Loops • 1-3: GPU / PyTorch / BPE • 4-7: MoE / SFT / RLVR / CoT • 8-10: MCP / Harness^2 / A2A	**H100 / B200 · CUDA** HBM3e vs. On-Chip SRAM FlashAttention-3 Tiling CUDA · OpenAI Triton FP8 & Microscaling FP4 NVLink 5 & InfiniBand MFU > 45% Utilization Prefill vs. Decode BW Confidential GPU TEEs ⚠ ANTI-PATTERNS & TRAPS × HBM Bandwidth Wall × CUDA OOM Crashes × Silent SDC Bit-Flips × Straggler GPU Stall	**uv · PyTorch · vLLM** brew · uv · conda · pip · venv PyTorch 2.x · JAX · SGLang torch.compile & XLA PagedAttention KV Cache uv.lock Reproducibility E2B / Firecracker VM Bubblewrap / Seatbelt TTFT / TPOT Async SLAs ⚠ ANTI-PATTERNS & TRAPS × CUDA ABI & Wheel Hell × Unsandboxed Host RCE × KV-Cache Preemption × pip Breaking flash-attn	**FineWeb · Byte-BPE** FineWeb-Edu · Dolma · Stack Byte-BPE · Tiktoken · Dedup Datatrove · NeMo Curator Verified Code/Math Synth 13-Gram Decontamination PII & Secret Scrubbing 128k-256k Vocab Size Curriculum Annealing ⚠ ANTI-PATTERNS & TRAPS × Synthetic Model Collapse × Test-Set Contamination × Tokenizer Glitch Tokens × PII Baked into Weights	**Sparse MoE · MLA** Sparse MoE · GQA / MLA Megatron-LM · TorchTitan Llama 4 · Qwen 3 · V3 Mamba / DeltaNet Hybrid RMSNorm, SwiGLU & RoPE 3D Parallel (DP/TP/PP/EP) Chinchilla Scaling Laws Perplexity (BPB) & NIAH ⚠ ANTI-PATTERNS & TRAPS × Loss Spikes & NaN Crash × MoE Routing Collapse × Pipeline Bubble Stall × RoPE Length Collapse	**ChatML · TRL · LoRA** TRL · Axolotl · Unsloth LoRA · QLoRA · DoRA PEFT Completion Loss Masking Multipack + Position IDs Tool & Trajectory SFT Cold-Start CoT Traces MergeKit (SLERP / DARE) IFEval Format Audits ⚠ ANTI-PATTERNS & TRAPS × Unmasked Prompt Loss × EOS/BOS Template Loops × Packed Attention Bleed × Catastrophic Forgetting
ORGANISM · KNOWLEDGE **RAG** Hybrid Vector, Lexical & Graph Retrieval-Augmented Generation External Grounding, Dense + BM25 Search, GraphRAG & Citation Verification • 1-3: SIMD / FAISS / Docling • 4-7: BGE-M3 / Rerank / HyDE • 8-10: Qdrant / Agentic RAG	**cuVS · SIMD · NVMe** NVIDIA cuVS / RAFT GPU AVX-512 Distance SIMD CAGRA GPU Graph Index NVMe Gen5 DiskANN I/O Int8 / Binary PQ Codes RDMA Index Sharding Sub-10ms p99 Vector SLA DRAM + SSD Tiered Pool ⚠ ANTI-PATTERNS & TRAPS × SSD Random-Read Thrash × HNSW Exceeding DRAM × PCIe Index Bottleneck × NUMA Socket Latency	**FAISS · Arrow · PQ** FAISS · HNSWlib · LanceDB psycopg3 · qdrant-client PyArrow Zero-Copy RAM BM25 · Tantivy · DuckDB Parquet Columnar Store Async gRPC / HTTP Pools ONNX / TRT Embed Serve mmap Vector Segments ⚠ ANTI-PATTERNS & TRAPS × Python GIL Chunk Lock × Pickle Index RCE Exploit × DB Pool Exhaustion × Pandas Copy RAM Blowup	**Docling · Chunking** Docling · Marker PDF ETL Contextual & AST Chunking Anthropic Chunk Headers Table-to-Markdown Parse Hard-Negative Mining Metadata & RBAC Tags CDC Incremental Sync Golden QA Eval Sets ⚠ ANTI-PATTERNS & TRAPS × Mid-Table Naive Split × PDF Column Scramble × Stale Deleted Vectors × Missing ACL Metadata	**BGE-M3 · ColBERTv2** Bi-Encoder InfoNCE Loss BGE-M3 · NV-Embed · ColBERT Matryoshka (MRL) Dims SPLADE Sparse Lexical Cross-GPU In-Batch Negs Bidirectional Encoders Multilingual Alignment MTEB & NDCG@10 Evals ⚠ ANTI-PATTERNS & TRAPS × Embedding Anisotropy × False-Negative Batches × Domain Acronym Blindness × 512-Token Truncation	**Rerank · RAFT SFT** Cross-Encoder bge-reranker Citation-Grounded RAFT SFT Domain Embed Fine-Tune Distractor Doc Immunity Query-Rewrite SFT Text-to-SQL / JSON SFT Margin-MSE Distillation 'I Don't Know' Abstain ⚠ ANTI-PATTERNS & TRAPS × Memorizing Stale Facts × Believing Distractors × Fake [1] Citation Tags × Narrow Query Overfit
ORGANISM · SENSORY **Multimodal** Vision, Audio, Speech, Video & Native Omni-Modal Models ViT Patch Perception, Native Speech Codecs, Spatio-Temporal Video & Cross-Modal Omni • 1-3: NVDEC / FFmpeg / Pairs • 4-7: SigLIP / LLaVA / Visual CoT • 8-10: Token Merge / LiveKit	**NVDEC · ViT · DSP** Hardware NVDEC / NVENC ViT Patch SRAM Tiling Apple M4 Unified RAM JPEG/HEVC HW Decode 2D/3D DiT Conv Kernels GPUDirect Video Ingest Sub-200ms Voice DSP 4K Frame BW Budgeting ⚠ ANTI-PATTERNS & TRAPS × CPU Video Decode Stall × 4K Frame PCIe Flood × Audio Sample Clock Drift × Video Thermal Throttle	**FFmpeg · OpenCV** torchvision · torchaudio LiveKit · whisper.cpp FFmpeg · PyAV · Decord NVIDIA DALI GPU Loader Pillow-SIMD · librosa vLLM Multimodal Serve Silero VAD Turn Runtime WebRTC Jitter Buffers ⚠ ANTI-PATTERNS & TRAPS × OpenCV BGR vs RGB Bug × FFmpeg Zombie Leaks × Audio Buffer Underruns × Pillow Bomb DoS Crash	**DataComp · OCR** DataComp · OBELICS Pairs OCR, Chart & Speech Sets CLIPScore Aesthetic Gate ShareGPT4V Recaptioning Scene-Cut Video Clips EnCodec / Mimi Tokens Face Blur & C2PA Scrub Set-of-Mark BBox Labels ⚠ ANTI-PATTERNS & TRAPS × Mismatched Alt-Text × Low-DPI OCR Misreads × Demographic Bias Skew × Audio Crosstalk Bleed	**SigLIP · Omni · DiT** SigLIP-2 · DINOv2 Vision Early-Fusion Omni & DiT Qwen2.5-VL · InternVL3 MLP / Q-Former Projector 3D Video RoPE Position RVQ Audio Codecs (Mimi) Flow-Matching Backbones Zero-Shot VQA & ASR ⚠ ANTI-PATTERNS & TRAPS × Modality Text-IQ Drop × Bag-of-Patches Blindness × 4K Quadratic ViT Blowup × Small-Text Patch Blur	**LLaVA · AnyRes SFT** Visual SFT (LLaVA / AnyRes) GUI BBox & Duplex Voice SFT Projector + LLM Stage-2 High-Res Chart/Doc SFT Interleaved Multi-Turn Set-of-Mark Visual SFT Moshi Full-Duplex SFT MMMU & ChartQA Evals ⚠ ANTI-PATTERNS & TRAPS × Blind Visual Guessing × Projector Detail Loss × BBox Coordinate Drift × Voice SFT Hurting Math
ORGANISM · ACTION **Embodied** GUI Computer Use, Physics World Models & VLA Robotics Browser/OS GUI Control, 3D Physics Simulators, World Models & VLA Robot Policies • 1-3: Edge NPU / Isaac / Teleop • 4-7: VLA π0 / ACT / Sim2Real • 8-10: 3DGS / OSWorld / Fleets	**Jetson Thor · PhysX** Edge NPU & Sensor ISP GPU PhysX / Warp Sim 1 kHz Real-Time Bus LiDAR / IMU Hardware EtherCAT / CAN-FD Bus INT4/FP8 Edge NPU 10,000x Parallel Sim Hardware E-Stop Relay ⚠ ANTI-PATTERNS & TRAPS × Edge Thermal Throttle × OS Jitter Breaking 1kHz × CSI Camera Frame Drop × Sim Contact Mismatch	**ROS 2 · Isaac · CDP** ROS 2 · Gymnasium · MuJoCo Playwright · LeRobot NVIDIA Isaac Lab / Gym CycloneDDS Middleware Headless Chromium Xvfb Open3D · Pinocchio IK Async Vectorized Envs Deterministic Replays ⚠ ANTI-PATTERNS & TRAPS × ROS 2 Multicast Storms × Zombie Browser Leaks × Euler Gimbal Lock Bug × Sim Step Drift Desync	**Open-X · GUI Traces** Open X-Embodiment · DROID Mind2Web · OSWorld · Ego4D ALOHA / VR Teleop Demos DOM + Screen + Action 3D Point-Cloud Align Joint-State Normalize RLDS / LeRobot Parquet Sim Domain Randomize ⚠ ANTI-PATTERNS & TRAPS × Teleop Hesitation Noise × Screen Resolution Drift × Action-Space Mismatch × Unlogged Confounders	**VLA π0 · V-JEPA 2** VLA (π0 · OpenVLA · RT-2) World Models (V-JEPA / Genie) Diffusion Action Policy Octo Multi-Embodiment Latent Next-Frame Pred 3D Point-Cloud Encoder SE(3) Equivariant Pose Zero-Shot Sim Transfer ⚠ ANTI-PATTERNS & TRAPS × Spurious Causal Confusion × World Physics Delusion × Mode-Averaging Freeze × Camera Viewpoint Fragile	**ACT · BC · OS-Atlas** Behavioral Cloning & ACT GUI Action SFT (OS-Atlas) k-Step Action Chunking Language-Conditioned BC Click / Type / Scroll SFT Wrist + 3rd-Person View New-Gripper LoRA Tune Offline Action MSE Eval ⚠ ANTI-PATTERNS & TRAPS × Compounding BC Drift × Copycat Action Inertia × Lighting Overfitting × Static Coordinate Click

ANATOMY	**6. Post-Train** Preference Alignment (RLHF/DPO) & Verifiable-Reward Reinforcement Learning (RLVR/GRPO) • Unit: Pairs, Rollouts & Advantages • Guide: Ch. 1.8, 3.2, 12.1-8, 30.2	**7. Test-Time** Test-Time Compute Scaling, System-2 Chain-of-Thought Reasoning & Tree Search • Unit: <think> Tokens, PRM & Trees • Guide: Ch. 13.1-8, 15.1-9, 30.2	**8. Context** Context Engineering, Four-Tier Memory, Tool Calling, MCP & Skills • Unit: Context, Memory & JSON-RPC • Guide: Ch. 17, 18.2-5, 22, 23	**9. Harness** Single-Agent Harness Engineering, Observe-Decide-Act Loops & Stateful Graph Workflows • Unit: State, Actions & Git Diffs • Guide: Ch. 16, 19, 20, 27	**10. Swarms** Multi-Agent Orchestration, Hierarchical Swarms, A2A Protocols & Shared State • Unit: A2A Tasks & Swarm State • Guide: Ch. 1.14, 14, 24, 26, 28, 30.10
LLM	**PPO · DPO · GRPO** PPO · DPO · KTO · SimPO RLVR: GRPO · DAPO · verl DeepSeek-R1 · o3 · QwQ Code/Math Test Oracles OpenRLHF · NeMo Aligner KL & Entropy Monitoring Deliberative Alignment Length-Controlled Win ⚠ ANTI-PATTERNS & TRAPS × Reward Hacking & Bloat × Policy Entropy Collapse × Zero-Variance Advantage × Judge Sycophancy Bias	**<think> CoT · MCTS** System-2 <think> CoT Best-of-N · PRM · MCTS o3 · Claude 4.6 · R1 Step-Level Process RM Adaptive Budget Forcing Exec-Filter Before Vote Pass@k vs. Majority@k AIME & ARC-AGI-2 Evals ⚠ ANTI-PATTERNS & TRAPS × 10x Overthinking Burn × PRM Verifier Hacking × Self-Correction Regress × Unfaithful Post-Hoc CoT	**MCP · 4-Tier Mem** Model Context Protocol 4-Tier Memory & SKILL.md Letta · Mem0 · Zep Episodic & Semantic Mem Progressive Skill Load Sleep-Time Consolidation /compact (<65% Window) MCP OAuth 2.1 & LoCoMo ⚠ ANTI-PATTERNS & TRAPS × Lost-in-the-Middle Rot × 100+ MCP Schema Bloat × SKILL.md / MCP Rug-Pull × Poisoned Memory State	**Claude Code · Loop** Claude Code · Codex · Cursor LangGraph · Harness^2 Loop CLAUDE.md / AGENTS.md Observe-Decide-Act Loop Typed State & Checkpoints Spec->Plan->Test->Verify Independent Test Grader SWE-bench & pass^k SLA ⚠ ANTI-PATTERNS & TRAPS × Deleting Failing Tests × Infinite Loop Token Burn × God-Node State Bloat × IDE Auto-Approve RCE	**A2A · OWC · AG-UI** Orchestrator–Worker–Critic Google A2A · CrewAI · AG2 Parallel Git Worktrees Red-vs-Blue Independent CopilotKit · AG-UI Agent Gateway & FinOps Cryptographic Agent KYA /.well-known/agent.json ⚠ ANTI-PATTERNS & TRAPS × Swarm Echo Hallucinate × $O(N^2)$ Chatty Token Burn × Confused Deputy Breach × Worktree Merge Conflict
RAG	**Search-R1 · DPO** Search-R1 Multi-Hop RLVR Citation Faithfulness DPO Query-Retrieve-Reason RL Exact-Match / F1 Oracles NLI Entailment Rewards Conflict Resolution RL Search Cost Penalty RL RAGTruth & FaithBench ⚠ ANTI-PATTERNS & TRAPS × Redundant Search Loops × Cherry-Picking Evidence × Over-Abstention Refusal × Uncited Extrapolation	**RRF · CRAG · HyDE** Hybrid RRF + Reranking IRCoT · CRAG · Self-RAG Query Decomposition HyDE Hypothetical Docs Top-100 -> 10 Reranker Interleaved Search + CoT Minimum-Span Extraction Dynamic Score Cutoffs ⚠ ANTI-PATTERNS & TRAPS × Infinite Rewrite Loops × HyDE Hallucination Drift × Reranker p99 SLA Blowup × Multi-Hop Error Cascade	**Qdrant · GraphRAG** Qdrant · Milvus · pgvector GraphRAG · LightRAG · Neo4j REFRAG / LLMLingua Comp Dense + BM25 Hybrid DB Leiden Graph Communities Parent-Doc Expansion Tenant RBAC Pre-Filter RAG Triad Faithfulness ⚠ ANTI-PATTERNS & TRAPS × Indirect Prompt Inject × Cross-Tenant Vector Leak × Context Stuffing Noise × GraphRAG Indexing Cost	**Agentic RAG Loop** Agentic RAG Workflows Plan–Search–Read–Verify Adaptive Router Gate Sub-Question DAG Merge Span-Level Citation Gate Source Freshness Check Web Fallback on KB Miss DeepResearch Benchmark ⚠ ANTI-PATTERNS & TRAPS × Rephrase Search Thrash × Silent Hallucination × Citation Laundering × Unbounded Research Lag	**Federated RAG MAS** Federated Knowledge Broker Parallel Deep-Research MAS SQL + Vector + Graph Bus Map-Reduce Lit Synthesis Fact-Checker vs. Writer OAuth On-Behalf-Of ACLs Shared Citation Board Cross-Silo Recall Evals ⚠ ANTI-PATTERNS & TRAPS × Circular Citation Loop × Cross-Dept ACL Leakage × Duplicate Doc Fetching × Dropping Minority Facts
Multimodal	**VLM RLVR · RLAIF-V** Visual Math & OCR RLVR RLAIF-V & Diffusion DPO IoU BBox Reward Oracles Chart/Table Verifier RL Flow-GRPO Aesthetic RL Voice Turn-Taking RLHF Visual Safety Alignment HallusionBench Audits ⚠ ANTI-PATTERNS & TRAPS × Text-Prior Blind Guess × Oversaturated Image Hack × Benign Image Refusals × Voice Barge-In Jitter	**Visual CoT · Zoom** Iterative Crop/Zoom CoT Active ROI Scan & Verify o3-Vision Crop Inspect OpenCV Code-Exec Zoom Video Keyframe Search Diffusion CFG Scaling Audio Re-Listen Verify Multi-Crop Voting ⚠ ANTI-PATTERNS & TRAPS × Zooming Irrelevant Bg × Blurry Crop Hallucinate × High-Res Token Overflow × CFG Over-Step Artifacts	**ToMe · Video KV** ViT Token Merge (PruMerge) Streaming Video KV Buffer Drop 75% Bg ViT Tokens Cross-Modal CLIP Memory Audio + Text Dual Index MCP Presigned Media URI Duplex Barge-In State Video-MME Long Context ⚠ ANTI-PATTERNS & TRAPS × 10m Video Filling 128k × Pruning Tiny OCR Text × Base64 JSON-RPC Crash × Audio-Video Sync Drift	**LiveKit · Pipecat** LiveKit / Pipecat Voice Render–Inspect–Fix Loop VAD + Barge-In Machine VLM UI Screenshot Diff ComfyUI Gen-Critique ControlNet Inpaint Loop C2PA Media Provenance WER & Turn-Latency SLA ⚠ ANTI-PATTERNS & TRAPS × Talking Over User Voice × Transcribing Own TTS × VLM Breaking CSS Layout × >3s Cascaded Voice Lag	**Multi-Modal Swarm** Cross-Modal Media Swarm Director–Video–Audio MAS Voice + Tool Subagents Character Style Board Scene + Foley Sync Gen VLM Visual QA Gatekeeper Streaming AG-UI Canvas Copyright & Brand Gate ⚠ ANTI-PATTERNS & TRAPS × Character Identity Drift × Voice Promising Failed Op × Audio-Video Lip Desync × Runaway Render Spend
Embodied	**Sim2Real · RLVR** OSWorld / WebArena RLVR Sim-to-Real PPO & HIL-SERL DOM / DB State Oracles Human-in-Loop SERL RL Friction/Mass Randomize Residual VLA Policy RL Control-Barrier Shield Docker Rollout Resets ⚠ ANTI-PATTERNS & TRAPS × Simulator Physics Exploit × Hardware Damage in RL × Reward-Hacking 'Buy' Btn × Sparse Credit Stall	**MPC · Tree Rollout** Model-Predictive Control World-Model Tree Search CEM / MPPI Sim Rollouts VLM Failure Replanning Pre-Click Hover Verify VoxPoser 3D Value Maps Receding-Horizon Exec Safety-Filter Critic ⚠ ANTI-PATTERNS & TRAPS × Planning Lag > 1kHz Loop × 5s Rollout Physics Drift × Unverified Destructive Click × Obstacle Deadlock Loop	**3DGS · DOM AXTree** 3D Scene Graphs & 3DGS DOM & AXTree Pruning OctoMap Occupancy Grid Set-of-Mark Element IDs Proprioceptive Ring Buf Kalman 6-DoF Object Track MCP Robot/Browser Bus Spatial Consistency QA ⚠ ANTI-PATTERNS & TRAPS × 50k-Node Raw DOM Flood × Dynamic Obstacle Ghosts × Out-of-Sight Forgetting × Odometry Frame Drift	**Computer Use Loop** Operator / Claude CU Loop System-2 VLM + System-1 VLA Screen -> Action -> Verify Pop-Up & Auth Recovery 2Hz Planner + 50Hz Motor Force/Torque Reflex Stop Checkout/Delete Gate OSWorld / WebArena Eval ⚠ ANTI-PATTERNS & TRAPS × Fake Pop-Up Injection × Infinite 'Pay' Clicks × UI Lag Misaligned Click × Ignoring Gripper Slip	**Fleet MAPF Swarm** Multi-Robot Fleet Control Digital-Twin + Physical MAS Conflict-Free MAPF Path Drone + Rover + Arm MAS V2X Spatial Perception Cloud Plan + Edge Reflex Human Teleop Handoff Fleet Deadlock Evals ⚠ ANTI-PATTERNS & TRAPS × Corridor Traffic Gridlock × Faulty Sensor Map Poison × Wi-Fi Drop Fleet Split × Dual-Arm Grasp Collision